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ORIGINAL RESEARCH

A Novel Approach for Biclustering Bipartite Networks: An Extension of Finite Mixtures of Latent Trait Analyzers

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Abstract
In the context of network data, bipartite networks are of particular interest, as they provide a useful description of systems representing relationships between sending and receiving nodes. In this framework, we extend the mixture of latent trait analyzers (MLTA) model with concomitant variables (nodal attributes) to perform a joint clustering of the two disjoint sets of nodes of a bipartite network, as in the biclustering framework. In detail, sending nodes are partitioned into clusters (called components) via a finite mixture of latent trait models. In each component, receiving nodes are partitioned into clusters (called segments) by adopting a flexible and parsimonious specification of the linear predictor. Residual dependence between receiving nodes is modeled via a multidimensional latent trait, as in the original MLTA specification. Furthermore, by incorporating nodal attributes into the model's latent layer, we gain insight into how these attributes impact the formation of components. To estimate model parameters, an EM-type algorithm based on a Gauss-Hermite approximation of intractable integrals is proposed. A simulation study is conducted to test the performance of the model in terms of clustering and parameters' recovery. The proposed model is applied to a bipartite network on pediatric patients possibly affected by appendicitis with the objective of identifying groups of patients (sending nodes) being similar with respect to subsets of clinical conditions (receiving nodes).

Keywords Network data · Model-based clustering · Co-clustering · EM algorithm · Gauss-Hermite quadrature

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1 Introduction

Network data analysis is particularly relevant when the research interest lies in analyzing relationships between units. In this context, bipartite networks are gaining increasing importance. These represent connections between two disjoint sets of nodes, called sending and receiving nodes, respectively. Note that the terms "sending" and "receiving" are simply used to distinguish the two different sets and do not directly refer to the direction of connections between nodes. The particular feature of this type of network is that connections are only allowed between nodes belonging to different sets, as shown in Fig. 1.

This peculiar type of network is useful for representing various phenomena, such as the boards of directors of companies (Davis et al., 2003), in which one group of nodes is represented by company directors, while their boards represent the other group. Another example in the economic context is given by the connections between countries and the products they export, which can be used to measure the complexity of a country economy (Hidalgo and Hausmann, 2009). On the other hand, an example in the biological field is given by the bipartite network of metabolic reactions, where the two sets of nodes represent metabolites and reactions to which they participate. In addition, researchers may be interested in exploring the links between genes and protein products to identify disease-specific susceptibility determinants (Pavlopoulos et al., 2018). Finally, in biomedical research, bipartite networks are used to represent potential drug-drug interactions and to analyze or predict unknown interactions (Guimerà and Sales-Pardo, 2013). Bipartite networks are also widely applied in social and behavioral research, with the aim of analyzing, for example, the participation of individuals to a set of social events (Davis et al., 1941) or the behaviors adopted by patients to avoid infection during a pandemic (Failli et al., 2024a).

In network data analysis, a relevant research objective frequently concerns the identification of clusters of nodes sharing similar characteristics, for dimensionality reduction purposes. This goal is frequently achieved using finite mixture specifications (McLachlan and Peel, 2000). Firstly introduced by Holland et al. (1983) and subsequently extended by, e.g., Snijders and Nowicki (1997); Nowicki and Snijders (2001); Daudin et al. (2008); Matias and Miele (2017); Bartolucci et al. (2018); Marino and Pandolfi (2022), stochastic block models (SBMs) represent a particular type of finite mixture allowing to identify clusters of nodes

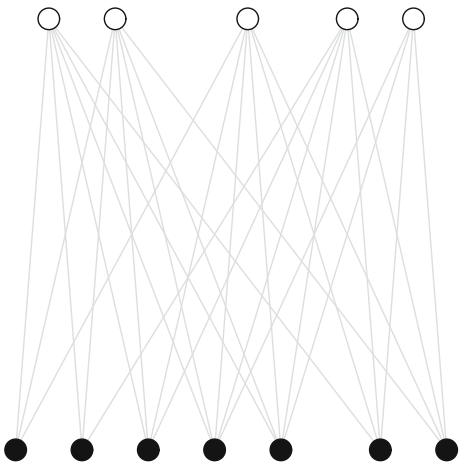


Fig. 1 Example of bipartite network

sharing similar relational profiles. The latent position cluster model introduced by Handcock et al. (2007) represents a further alternative, by allowing the identification of communities of strongly connected nodes. This is done by considering a finite mixture of node-specific latent positions such that the probability of a tie between a pair of nodes decreases as the distance between positions increases. Furthermore, Vu et al. (2013) employ a finite mixture approach for clustering discrete-valued networks using scalable exponential family models. In the specific context of bipartite networks, Aitkin et al. (2014) and Aitkin et al. (2017) consider a latent class model to identify groups of sending nodes sharing similar (unobserved) features.

However, we can observe that traditional clustering methods typically focus on identifying similarities among sending nodes based on their connections to receiving nodes, which can be limiting. In fact, sending nodes may exhibit similar behavior with respect to a specific subset of receiving nodes, while showing little or no similarity with others. In such cases, the analysis could benefit from a joint clustering of both sending and receiving nodes, as it is commonly done in data matrices where subsets of rows (representing units) show similar patterns with specific subsets of columns (representing variables). This approach allows to capture more complex and nuanced structures within the bipartite network. To this aim, a biclustering approach (also known in the literature as co-clustering, block clustering, simultaneous clustering, or block modeling) may be used and, also in this framework, finite mixtures play a relevant role. Outside of the network data framework , Govaert and Nadif (2003) and Keribin et al. (2014) exploit a block mixture model to perform a simultaneous clustering of units and categorical variables, while Martella et al. (2008) propose an extension of the mixture of factor analyzers model (Ghahramani and Hinton, 1997) for biclustering continuous data. Martella and Alfò (2017) consider a finite mixture specification for the joint clustering of units and variables having a discrete nature; Alfò et al. (2024) further extend this latter approach to deal with discrete longitudinal data. In the context of bipartite networks, Wyse and Friel (2012) and Wyse et al. (2017) employ the latent blockmodel (LBM) introduced by Govaert and Nadif (2008) to perform a simultaneous clustering of the two sets of nodes in a bipartite network.

In this paper, we extend the mixture of latent trait analyzers (MLTA) model, introduced by Gollini and Murphy (2014) for multivariate categorical (binary) data, in a biclustering perspective. This originally combines features of latent class and latent trait models, with the twofold objective of clustering sending nodes via a finite mixture specification and modeling the residual dependence between receiving nodes via a continuous multidimensional latent trait. In the applied literature, the MLTA is used by Gollini (2020) to identify groups of terrorists based on the attacks to which they participate; Failli et al. (2024a) employ this model specification to identify groups of patients exhibiting similar behaviors in the adoption of preventive measures against COVID-19 infection, while Failli et al. (2024b) apply the MLTA to study disparities of European residents with respect to various digital competences. In all these cases, a clustering of sending nodes only is obtained.

Our proposal is to modify the MLTA specification by allowing for a joint clustering of sending and receiving nodes; in other terms, the MLTA model is extended in a biclustering perspective. If applied to the terrorist-attack bipartite network, this extension would enable the identification of homogeneous subsets of events based on the groups of terrorists involved. Similarly, for the bipartite network analyzed by Failli et al. (2024a), adopting a biclustering perspective would facilitate the discovery of homogeneous subsets of preventive measures that are similarly widespread across specific subsets of patients.

We start from the MLTA model proposed by Gollini and Murphy (2014), which provides a partitioning of sending nodes into subsets called components. Then, within each component,

we allow for a partitioning of sending nodes into subsets called segments by following an approach similar to that detailed by Martella and Alfò (2017) and Alfò et al. (2024). This is based on a flexible and parsimonious specification of the model linear predictor. Furthermore, according to Failli et al. (2024a), we allow for the inclusion of concomitant variables (nodal attributes) on the latent layer of the model to detect how they influence component formation.

The performance of the model in terms of parameters' recovery, uncertainty evaluation, and clustering is evaluated through a large-scale simulation study based on a different number of nodes and partitions. The model selection performance is also evaluated. Furthermore, the proposed approach is applied to a bipartite network of pediatric patients with potential appendicitis (Marcinkevičs et al., 2023). Patients correspond to sending nodes, while their clinical conditions correspond to the receiving ones. A link between the two does exist if the given patient manifests the given clinical condition. The aim is twofold: (i) identifying groups of patients sharing specific subsets of clinical conditions and (ii) taking into account how patients' characteristics influence the probability of belonging to a particular component.

The paper is organized as follows. In Section 2, we briefly describe the assumptions underlying the original specification of the MLTA model for bipartite networks. In Section 3, we extend the MLTA model in a biclustering perspective, also describing parameter estimation and model selection. Section 4 shows the results of the simulation study, while Section 5 presents the application of both the original MLTA and the proposed extension to the bipartite network of pediatric patients. Section 6 contains concluding remarks and details of further extensions of the approach.

2 MLTA Model

Let Y_{ik} denote a binary random variable and y_{ik} the corresponding observed value for the i-th sending node, $i = 1, \ldots, N$, and the k-th receiving node, $k = 1, \ldots, R$. A bipartite network can be formally described by a random incidence matrix $\boldsymbol{Y} = \{Y_{ik}\}$, with elements

$$Y_{ik} = \begin{cases} 1 & \text{if sending node } i \text{ is connected with receiving node } k, \\ 0 & \text{otherwise.} \end{cases}$$

The MLTA model (Gollini and Murphy, 2014; Gollini, 2020) combines latent class (Lazarsfeld and Henry, 1968; Goodman, 1974) and latent trait analysis (Bartholomew et al., 2011) by assuming that the set of N sending nodes can be divided into G distinct components and that the propensity for a sending node to be connected with a given receiving node depends also on a multidimensional, sending-specific, continuous, latent trait. Formally, it is assumed that every sending node belongs to one out of G unobserved components identified by a discrete latent variable $\boldsymbol{z}_i = (z_{i1}, \ldots, z_{iG})' \overset{iid}{\sim} \text{Multinomial}(1, (\eta_1, \ldots, \eta_G)')$, whose generic element is

$$z_{ig} = \begin{cases} 1 & \text{if sending node } i \text{ belongs to component } g, \\ 0 & \text{otherwise.} \end{cases}$$

The parameter η_g denotes the probability that a randomly selected sending node belongs to component g, under the constraints that $\sum_{g=1}^{G} \eta_g = 1$ and $\eta_g > 0$, $g = 1, \ldots, G$.

Furthermore, the model assumes the existence of a D-dimensional, sending-specific, continuous, latent trait $\boldsymbol{u}_i$ distributed according to a D-variate standard Gaussian density, i.e., $\boldsymbol{u}_i \sim \mathcal{N}_D(\boldsymbol{0}, \boldsymbol{I})$. This is assumed to capture the (potential) residual heterogeneity of sending nodes in the way they connect to receiving nodes. A practical example can be found in the

network analyzed by Failli et al. (2024a) and referenced in the previous section, where the sending nodes-specific latent trait can be interpreted as the residual unobserved propensity of each individual (sending node) to adopt specific measures (receiving nodes) to prevent COVID-19 infection. Also note that constraints on the distribution of $\boldsymbol{u}_i$ (specifically, the zero mean vector and the identity covariance matrix) are necessary to ensure identifiability of the model (Bartholomew et al., 2011). As far as Gaussianity, this is a rather standard assumption in latent trait analysis, driven by both mathematical and computational convenience, as well as by the fact that many real-world phenomena can be adequately described using this distribution.

A local independence assumption is also considered; that is, conditional on $\boldsymbol{z}_i$ and $\boldsymbol{u}_i$, response variables in the vector $\boldsymbol{Y}_i = (Y_{i1}, \ldots, Y_{iR})'$ are assumed to be independent Bernoulli random variables with parameters $\pi_{gk}(\boldsymbol{u}_i)$, $k = 1, \ldots, R$, modeled through the following logistic function:

$$\pi_{gk}(\boldsymbol{u}_i) = \Pr(Y_{ik} = 1 \mid \boldsymbol{u}_i, z_{ig} = 1) = \frac{1}{1 + \exp[-(b_{gk} + \boldsymbol{w}'_{gk}\boldsymbol{u}_i)]}. \tag{1}$$

Here, the model intercept b_{gk} is a component-specific latent effect measuring the attractiveness of the k-th receiving node for sending nodes belonging to the g-th component. Furthermore, the slope $\boldsymbol{w}_{gk}$ associated with the latent variable $\boldsymbol{u}_i$ is meant to capture the influence of the D-dimensional latent trait on the probability of a connection between the k-th receiving node and the sending nodes belonging to the g-th component. Statistically significant estimates of the influence parameter $\boldsymbol{w}_{gk}$ indicate association between receiving nodes, as well as the presence of potential residual heterogeneity between sending nodes in the g-th component in the way they connect to the k-th receiving node, with respect to the baseline level provided by b_{gk}. Last, the discrete latent variable $\boldsymbol{z}_i$ and the continuous latent trait $\boldsymbol{u}_i$ are assumed to be independent.

3 Extending the MLTA Model for Biclustering Bipartite Networks

As it is pointed out in Section 1, our proposal aims at performing a joint clustering of sending and receiving nodes. The former are partitioned into clusters called components, while the latter are partitioned into clusters called segments. This is done by following an approach similar to that proposed by Martella and Alfò (2017) and extended to the longitudinal framework by Alfò et al. (2024). In detail, the logistic function in Eq. 1 is modified as

$$\pi_{gk}(\boldsymbol{u}_i) = \Pr(Y_{ik} = 1 \mid \boldsymbol{u}_i, z_{ig} = 1) = \frac{1}{1 + \exp[-(b_g + \boldsymbol{a}'_{gk}(\boldsymbol{\mu} + \boldsymbol{u}_i))]}. \tag{2}$$

Here, the model intercept b_g is a component-specific latent effect that is meant to capture the baseline tendency of sending nodes in the g-th component to connect to receiving nodes. On the other side, $\boldsymbol{\mu}$ is a D-dimensional vector of fixed effects which is assumed to be constant across components, while $\boldsymbol{u}_i \sim \mathcal{N}_D(\boldsymbol{0}, \boldsymbol{I})$ is a D-dimensional, continuous, latent trait associated to the i-th sending node. Last, $\boldsymbol{a}_{gk}$ is a D-dimensional row stochastic vector ($D \leq R$) with elements

$$a_{gkd} = \begin{cases} 1 & \text{if, within the } g\text{-th component, receiving node } k \text{ belongs to segment } d, \\ 0 & \text{otherwise.} \end{cases}$$

According to this definition, the vector $\boldsymbol{a}_{gk}$ allows to select a single term only from both $\boldsymbol{\mu}$ and $\boldsymbol{u}_i$. Thus, for a sending node i belonging to the g-th component ($z_{ig} = 1$) and a receiving node belonging to the d-th segment ($a_{gkd} = 1$), the connection probability above becomes

$$\pi_{gk}(u_{id}) = \frac{1}{1 + \exp[-(b_g + \mu_d + u_{id})]}.$$

From this expression, it is evident that μ_d is a fixed effect that increases or decreases the attractiveness of receiving nodes in the d-th segment with respect to the general tendency dictated, for sending nodes in the g-th component, by b_g. On the other side, u_{id} is a sending- and segment-specific latent trait capturing the residual heterogeneity of the i-th sending node in the way it connects to the receiving nodes in the d-th segment.

When comparing the linear predictor postulated under the original MLTA framework with the proposed modified version, i.e., when comparing Eqs. 2 and 1, we may notice that the component- and receiving-specific intercept b_{gk} under the former formulation is now constrained to be constant across k. That is, we impose the constraint $b_{gk} = b_g$, for all $k = 1, \ldots, R$. This restriction is important from a biclustering perspective: in conjunction with the other parameters in the linear predictor, it allows for the joint clustering of rows and columns of the incidence matrix. Indeed, under the proposed approach, each bicluster is identified by combining the component- and segment-specific fixed effects b_g and μ_d, respectively. Then, the latent trait captures the residual heterogeneity specific of each sending node in terms of connections with those receiving nodes belonging to a specific segment.

Last, following the strategy adopted by Failli et al. (2024a), we model the effect that nodal attributes (i.e., observed characteristics of sending nodes) may have on component membership. This goal is achieved by letting the component probabilities η_g vary across sending nodes via the following latent class regression model:

$$\eta(\boldsymbol{x}_i; \boldsymbol{\beta}_g) = \Pr(z_{ig} = 1 \mid \boldsymbol{x}_i; \boldsymbol{\beta}_g) = \frac{\exp\{\boldsymbol{x}_i' \boldsymbol{\beta}_g\}}{1 + \sum_{g'=2}^{G} \exp\{\boldsymbol{x}_i' \boldsymbol{\beta}_{g'}\}}, \quad g = 2, \ldots, G.$$

Here, $\boldsymbol{\beta}_g = (\beta_{0g}, \beta_{1g}, \ldots, \beta_{Jg})'$ denotes a $(J + 1)$-dimensional vector of coefficients measuring the impact of nodal attributes $\boldsymbol{x}_i = (1, x_{i1}, \ldots, x_{iJ})'$ on prior component probabilities. We can see that the original MLTA specification by Gollini and Murphy (2014) is a special case of the proposed model in which the slopes $(\beta_{1g}, \ldots, \beta_{Jg})'$ are all set to zero.

An overall understanding of the proposal can be obtained by looking at Fig. 2, where the path diagram representing how the different model parts interact is presented.

To conclude, it is worth emphasizing that the proposed specification allows for a component-specific partitioning of receiving nodes. This represents a clear step ahead when compared to the LBM approach by Wyse and Friel (2012) and Wyse et al. (2017), relying

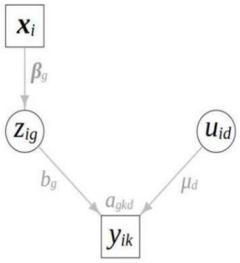


Fig. 2 Path diagram for the proposed extended MLTA model

instead on a rigid grid that produces a fixed partitioning of sending and receiving nodes. The proposed approach thus ensures a greater flexibility, by also allowing for a different number of segments in each mixture component. This makes the model able to capture in a natural manner underlying patterns in the data, which may be specific to some subsets of sending nodes only. For example, in a gene-patient bipartite network, certain groups of genes may exhibit similar expression patterns among healthy patients, while showing distinct expression patterns across patients affected by different diseases. Clearly, the enhanced flexibility comes at the cost of a higher complexity of the model, which however remains always under control.

3.1 Parameter Estimation

Let $\boldsymbol{\theta} = (\boldsymbol{\beta}_2, \ldots, \boldsymbol{\beta}_G, b_1, \ldots, b_G, \boldsymbol{a}_{11}, \ldots, \boldsymbol{a}_{GR}, \mu_1, \ldots, \mu_D)'$ represent the S-dimensional vector of all free model parameters. Given the assumptions described in the previous section, the likelihood function of the model can be written as

$$L(\boldsymbol{\theta}) = \prod_{i=1}^{N} L_i(\boldsymbol{\theta}) = \prod_{i=1}^{N} \left(\sum_{g=1}^{G} \eta(\boldsymbol{x}_i; \boldsymbol{\beta}_g) \int_{\mathbb{R}^D} f(\boldsymbol{y}_i \mid \boldsymbol{u}_i, z_{ig} = 1) f(\boldsymbol{u}_i) d\boldsymbol{u}_i \right),$$

where $f(\boldsymbol{y}_i \mid \boldsymbol{u}_i, z_{ig} = 1)$ denotes the product of R Bernoulli probability mass functions and the dependence from model parameters b_g, $\boldsymbol{\mu}$, and $\boldsymbol{a}_{g1}, \ldots, \boldsymbol{a}_{gR}$, is suppressed to ease the notation. In the present framework, the complete-data log-likelihood function is defined as

$$\ell_c(\boldsymbol{\theta}) = \sum_{i=1}^{N} \sum_{g=1}^{G} z_{ig} \log[f(\boldsymbol{y}_i \mid \boldsymbol{u}_i, z_{ig} = 1)] + \sum_{i=1}^{N} \sum_{g=1}^{G} z_{ig} \log[\eta(\boldsymbol{x}_i; \boldsymbol{\beta}_g)] + \sum_{i=1}^{N} \log[f(\boldsymbol{u}_i)]. \tag{3}$$

At the $(t+1)$-th iteration of the Expectation-Maximization (EM) algorithm (Dempster et al., 1977), the E-step consists in computing the expectation of Eq. 3, conditional on the observed data and the current parameter estimates $\boldsymbol{\theta}^{(t)}$. That is, we need to compute

$$\begin{aligned} Q(\boldsymbol{\theta} \mid \boldsymbol{\theta}^{(t)}) = & \sum_{i=1}^{N} \sum_{g=1}^{G} \int_{\mathbb{R}^D} \hat{z}_{ig}^{(t+1)} f(\boldsymbol{u}_i \mid z_{ig} = 1, \boldsymbol{y}_i, \boldsymbol{\theta}^{(t)}) \log[f(\boldsymbol{y}_i \mid \boldsymbol{u}_i, z_{ig} = 1)] \, d\boldsymbol{u}_i + \\ & + \sum_{i=1}^{N} \sum_{g=1}^{G} \hat{z}_{ig}^{(t+1)} \log[\eta(\boldsymbol{x}_i; \boldsymbol{\beta}_g^{(t)})] + \sum_{i=1}^{N} \int_{\mathbb{R}^D} f(\boldsymbol{u}_i \mid \boldsymbol{y}_i, \boldsymbol{\theta}^{(t)}) \log[f(\boldsymbol{u}_i)] \, d\boldsymbol{u}_i. \end{aligned} \tag{4}$$

In the above expression, $f(\boldsymbol{u}_i \mid z_{ig} = 1, \boldsymbol{y}_i, \boldsymbol{\theta}^{(t)})$ denotes the posterior density of $\boldsymbol{u}_i$, conditional on sending node i belonging to component g, the current parameter estimates, and the observed data, $f(\boldsymbol{u}_i \mid \boldsymbol{y}_i, \boldsymbol{\theta}^{(t)})$ is the posterior density of $\boldsymbol{u}_i$, conditional on the current parameter estimates and the observed data only, and, last, $\hat{z}_{ig}^{(t+1)}$ is the posterior probability for sending node i to belong to the g-th component. This latter quantity can be computed as

$$\hat{z}_{ig}^{(t+1)} = \Pr(z_{ig} = 1 \mid \boldsymbol{y}_i, \boldsymbol{\theta}^{(t)}) = \frac{\eta(\boldsymbol{x}_i; \boldsymbol{\beta}_g) \int_{\mathbb{R}^D} f(\boldsymbol{y}_i \mid \boldsymbol{u}_i, z_{ig} = 1) f(\boldsymbol{u}_i) d\boldsymbol{u}_i}{\sum_{g=1}^{G} \eta(\boldsymbol{x}_i; \boldsymbol{\beta}_g) \int_{\mathbb{R}^D} f(\boldsymbol{y}_i \mid \boldsymbol{u}_i, z_{ig} = 1) f(\boldsymbol{u}_i) d\boldsymbol{u}_i}. \tag{5}$$

The M-step of the algorithm consists in updating the model parameters by maximizing the expected complete-data log-likelihood function in Eq. 4 with respect to $\boldsymbol{\theta}$. In detail,

parameters b_g and $\boldsymbol{\mu}$ are estimated by finding the zeros of the expected score function, computed with respect to the posterior distribution of the indicator variables, z_{ig}'s, and the continuous latent traits, $\boldsymbol{u}_i$'s. As far as the vector $\boldsymbol{a}_{gk}$ is concerned, this is estimated, for fixed g and k, by identifying the segment d providing the maximum value for the expected complete-data log-likelihood. Finally, the multinomial logit coefficients, $\boldsymbol{\beta}_g$, are estimated by maximizing the likelihood of a multinomial logit model via a Newton-Raphson algorithm, with weights provided by the $\hat{z}_{ig}$'s derived at the E-step of the algorithm, as in Eq. 5. Prior probabilities $\eta(\boldsymbol{x}_i; \boldsymbol{\beta}_g)$ are updated accordingly.

The procedure is repeated until the following stopping rule is satisfied:

$$|| \ \ell_c(\hat{\boldsymbol{\theta}}^{(t+1)}) - \ell_c(\hat{\boldsymbol{\theta}}^{(t)}) \ || < \epsilon,$$

where $\epsilon > 0$ denotes a given tolerance level. In the following, we set $\epsilon = 10^{-4}$.

3.1.1 Approximating Intractable Integrals

As it is clear from the previous section, the EM algorithm requires the calculation of multi-dimensional integrals that do not have closed-form solutions. To address this issue, Gollini and Murphy (2014) suggest the use of a variational approximation. While computationally efficient and easy to implement, the variational approach approximates the log-likelihood function through a lower bound, often resulting in slow convergence of the EM algorithm and in a high sensitivity to parameter initialization. In contrast, Gauss-Hermite quadrature (see, e.g., Pinheiro and Bates, 1995 and Skrondal and Rabe-Hesketh, 2004) provides a more accurate approximation of the log-likelihood function, proving to be especially advantageous when working with low-dimensional integrals (latent traits). As in this context, our goal is to achieve dimensionality reduction of the original incidence matrix and the latent trait size directly corresponds to the number of segments (that is typically small), we opt for this latter approach.

The main idea behind Gaussian quadrature techniques is to approximate an integral through a weighted sum of the integrand function, evaluated on a fixed set of abscissas with given weights. For integrals of the form $\int_{\mathbb{R}^D} h(\boldsymbol{u}_i) e^{-||\boldsymbol{u}_i||^2} d\boldsymbol{u}_i$, the Gauss-Hermite quadrature leads to the following approximation:

$$\int_{\mathbb{R}^D} h(\boldsymbol{u}_i) e^{-||\boldsymbol{u}_i||^2} d\boldsymbol{u}_i \approx \sum_{q_1 \dots q_D} h(\boldsymbol{u}_{q_1 \dots q_D}) \prod_{l=1}^{D} \omega_{q_l}, \tag{6}$$

where

$$\omega_{q_l} = \frac{2^{Q-1} Q! \sqrt{\pi}}{Q^2 [H_{Q-1} \boldsymbol{u}_{q_l}]^2}.$$

Here, $\boldsymbol{u}_{q_1 \dots q_D} = (u_{q_1}, \dots, u_{q_D})'$ represents a Q-tuple defined by the Cartesian product of the roots of the Q-th order Gauss-Hermite polynomial $H_Q(\boldsymbol{u}_{q_l})$, for $q_l \in \{1, \dots, Q\}$ and $l \in \{1, \dots, D\}$. On the other side, ω_{q_l} are the corresponding quadrature weights, and $\sum_{q_1 \dots q_D}$ is used as a shorthand for $\sum_{q_1=1}^{Q} \cdots \sum_{q_D=1}^{Q}$.

Let us focus on solving integrals of the form

$$\zeta_{ig} = \int_{\mathbb{R}^D} f(\boldsymbol{y}_i \mid \boldsymbol{u}_i, z_{ig} = 1) f(\boldsymbol{u}_i) d\boldsymbol{u}_i, \tag{7}$$

required for deriving both the likelihood and the posterior probabilities $\hat{z}_{ig}$ in the EM algorithm; the computation follows a similar root when the expected score functions to estimate b_g,

$\boldsymbol{\mu}$, and $\boldsymbol{a}_{gk}$ need to be derived. As stated before, since the D-dimensional continuous latent trait $\boldsymbol{u}_i$ follows a D-variate standard Gaussian distribution, i.e., $f(\boldsymbol{u}_i) = (2\pi)^{-D/2} \exp\{-\frac{1}{2}\boldsymbol{u}_i'\boldsymbol{u}_i\}$, the integral in Eq. 7 can be rewritten as a function of variables $\tilde{\boldsymbol{u}}_i = \frac{\boldsymbol{u}_i}{\sqrt{2}}$; that is,

$$\zeta_{ig} \propto \sqrt{2^D} \int_{\mathbb{R}^D} f(\boldsymbol{y}_i \mid \sqrt{2}\tilde{\boldsymbol{u}}_i, z_{ig} = 1)\, e^{-||\tilde{\boldsymbol{u}}_i||^2} d\tilde{\boldsymbol{u}}_i.$$

This integral has the same form as Eq. 6, allowing the following approximation to be applied directly

$$\zeta_{ig} \approx \sqrt{2^D} \sum_{q_1 \ldots q_D} f(\boldsymbol{y}_i \mid \boldsymbol{u}^*_{q_1 \ldots q_D}, z_{ig} = 1) \prod_{l=1}^{D} \omega_{q_l},$$

with $\boldsymbol{u}^*_{q_1 \ldots q_D} = \sqrt{2}\boldsymbol{u}_{q_1 \ldots q_D}$.

3.1.2 Parameter Initialization and Clustering

As it is typically the case with latent variables, an initialization strategy based on multiple starting points can help the estimation algorithm not to get trapped in local maxima of the likelihood function. In detail, this can be run starting from different initial values, then retaining the solution providing the maximum (log-)likelihood value. In our proposal, the following initialization strategy is adopted. The $\boldsymbol{\beta}_g$ parameters are all initially set equal to zero, thus leading to uniform component probabilities $\eta(\boldsymbol{x}_i; \boldsymbol{\beta}_g)$, $g = 1, \ldots, G$, $i = 1, \ldots, N$. The parameters b_g and $\boldsymbol{\mu}$ are initialized for convenience via a random generation from a Gaussian distribution with zero mean and unit variance. Last, the vector $\boldsymbol{a}_{gk}$ is initialized, for each component g, by applying a k-means clustering method to the columns of the incidence matrix to speed up convergence, as suggested by Martella and Alfò (2017).

Once the algorithm halts, each sending node is assigned to the different components via a maximum a posteriori (MAP; Bishop, 2006) rule based on the last estimated posterior probabilities $\hat{z}_{ig}$'s, while each receiving node is automatically assigned to one of the segments according to the final estimate of the vector $\hat{\boldsymbol{a}}_{gk}$.

3.2 Standard Errors and Model Selection

Standard errors for the estimates obtained via the EM algorithm can be obtained following the method suggested by Louis (1982) and Oakes (1999) and by considering a sandwich formula (Royall, 1986; White, 1980). Denoting by $\hat{\boldsymbol{\theta}}$ the vector of parameter estimates, the corresponding standard errors are obtained as the square root of the diagonal elements of the following covariance matrix:

$$\widehat{\text{Cov}}(\hat{\boldsymbol{\theta}}) = \widehat{\boldsymbol{J}}(\hat{\boldsymbol{\theta}})^{-1} \widehat{\boldsymbol{K}}(\hat{\boldsymbol{\theta}}) \widehat{\boldsymbol{J}}(\hat{\boldsymbol{\theta}})^{-1}.$$

Here, $\widehat{\boldsymbol{J}}(\hat{\boldsymbol{\theta}})$ is the observed information matrix; its estimate is obtained by computing the first numerical derivative of the score vector $S(\boldsymbol{\theta}) = \partial \ell(\boldsymbol{\theta}) / \partial \boldsymbol{\theta}$, evaluated at $\hat{\boldsymbol{\theta}}$. Furthermore, $\widehat{\boldsymbol{K}}(\hat{\boldsymbol{\theta}})$ provides an estimate for the covariance matrix of the score vector and is computed as $\widehat{\boldsymbol{K}}(\hat{\boldsymbol{\theta}}) = \sum_{i=1}^{N} S_i(\hat{\boldsymbol{\theta}}) S(\hat{\boldsymbol{\theta}})'$, where $S_i(\hat{\boldsymbol{\theta}})$ is the individual contribution to the score function for the i-th sending node, evaluated at $\hat{\boldsymbol{\theta}}$.

As far as model selection is concerned, both the number of components (G) and the number of segments (D) are considered as fixed quantities. The model is estimated for different values

of G and D, and the optimal solution is selected as the one corresponding to the smallest value of the chosen information criterion; possible choices are the Bayesian information criterion (BIC; Schwarz, 1978):

$$\text{BIC} = -2\ell(\hat{\boldsymbol{\theta}}) + \nu \log(N),$$

where ν is the number of free parameters, or the integrated classification likelihood (ICL; Biernacki et al., 2000)

$$\text{ICL} = \text{BIC} - \sum_{g=1}^{G}\sum_{i=1}^{N} \hat{z}_{ig} \log(\hat{z}_{ig}).$$

In the former case, the optimal model corresponds to the minimum value of the criterion, while in the latter, this has to be maximized.

4 Simulation Study

The performance of the model in terms of parameters' recovery, clustering, and uncertainty evaluation is analyzed through a simulation study based on a different number of nodes, components, and segments, as described in Sections 4.1–4.4.

4.1 Simulation Setup

Twelve different scenarios based on $B = 100$ simulated samples are considered; these are based on a varying number of sending nodes ($N = 100, 500, 1000$), receiving nodes ($R = 20, 30$), components ($G = 3, 4$), and segments ($D = 2, 3$). As regards the latent class variable, we consider a single nodal attribute, i.e., $\boldsymbol{x}_i = (1, x_{i1})'$, with x_{i1} drawn from a Gaussian distribution with mean and variance both equal to 1. The latent structure is also defined by setting $\boldsymbol{\beta}_2 = (1, -0.4)'$ and $\boldsymbol{\beta}_3 = (1.5, -0.9)'$, for $G = 3$, and $\boldsymbol{\beta}_2 = (1, -0.4)'$, $\boldsymbol{\beta}_3 = (1.5, -0.9)'$, and $\boldsymbol{\beta}_4 = (2, -1.3)'$, for $G = 4$. Furthermore, we set $\boldsymbol{b} = (-1.7, 0, 1.7)'$ and $\boldsymbol{\mu} = (-2, 0.5)'$, when $G = 3$ and $D = 2$, and $\boldsymbol{b} = (-1.7, 0, 1.7, 0.7)'$ and $\boldsymbol{\mu} = (-2, 0.5, 1.5)'$, when $G = 4$ and $D = 3$. Furthermore, the multidimensional continuous latent trait is randomly drawn from a multivariate Gaussian distribution with zero mean vector and identity covariance matrix, i.e., $\boldsymbol{u}_i \sim \mathcal{N}_D(\boldsymbol{0}, \boldsymbol{I})$. In each scenario, the multi-start strategy described in Section 3.1.2, based on 100 random starts, is adopted.

The choice of this setup is motivated by the need to study the behavior of the proposed method when the network size increases and the biclustering structure becomes more complex. We expect that the ability of the proposed estimation method to recover the true partitions and the true parameter values improves with network size, while the performance is less satisfying with more complex structures.

4.2 Simulation Results: Clustering Recovery

The ability of the proposal in correctly classifying sending and receving nodes is evaluated through the adjusted Rand index (ARI; Hubert and Arabie, 1985), which is a standard measure of the agreement between the true and estimated partitions. Its expected value is zero in the case of random partition, while it is 1 in the case of perfect agreement between the two. Results of the simulation study are shown in Tables 1 and 2. Looking at these tables, we note that, as the number of sending and receiving nodes increases, the classification improves for both sending and receiving nodes. As expected, ARI values worsen as the number of partitions increases, as is evident when comparing Tables 1 and 2. In detail, when $G = 4$

Table 1 Mean (median) ARI across samples for $G = 3$, $D = 2$, and varying N and R

		$R = 20$		$R = 30$	
		Sending	Receiving	Sending	Receiving
N	100	0.71 (0.72)	0.83 (0.93)	0.82 (0.83)	0.91 (0.96)
	500	0.75 (0.75)	0.96 (1.00)	0.83 (0.83)	1.00 (1.00)
	1000	0.75 (0.77)	0.96 (1.00)	0.83 (0.83)	1.00 (1.00)

and $D = 3$, larger network sizes are needed to obtain a good clustering performance. We expect the classification ability of the model to further improve when considering bipartite networks with a larger number of nodes.

4.3 Simulation Results: Parameters' Recovery

Tables 3 and 4 show the absolute bias and mean squared error (MSE) values across samples for $\boldsymbol{b} = (b_1, \ldots, b_G)'$, $\boldsymbol{\mu} = (\mu_1, \ldots, \mu_D)'$, and $\boldsymbol{\beta} = (\boldsymbol{\beta}_2, \ldots, \boldsymbol{\beta}_G)'$, obtained when letting the number of nodes, components, and segments vary. Examining these tables, it becomes evident that as the network size increases, our ability to accurately identify the true values of the model parameters improves. Both the absolute bias and the MSE decrease, approaching zero as N and R grow larger. However, it is also clear that the estimation algorithm performs less effectively when dealing with more complex latent structures. Specifically, as the number of biclusters increases (reflected by larger values of G and D) a higher sample size is required to achieve satisfactory results.

4.4 Simulation Results: Standard Error Estimates and Model Selection

In this section, we describe the results of a reduced simulation study conducted to evaluate the performance of the proposed approach in measuring uncertainty of parameter estimates, along with the ability of the proposed model selection procedure in identifying the optimal number of components and segments. We simulate $B = 100$ networks, made by $N = 100$ sending nodes and $R = 20$ receiving nodes, by partitioning the former into $G = 3$ components and the latter into $D = 2$ segments, via the proposed model specification. Parameter values are set exactly as described in Section 4.1 for the setting based on the same number of components and segments. In detail, we set $\boldsymbol{\beta}_2 = (1, -0.4)'$, $\boldsymbol{\beta}_3 = (1.5, -0.9)'$, $\boldsymbol{b} = (-1.7, 0, 1.7)'$, and $\boldsymbol{\mu} = (-2, 0.5)'$. As before, the latent trait is randomly drawn from a D-variate standard Gaussian distribution. A multi-start strategy based on 100 random starts is adopted to avoid local-maxima solutions.

Table 5 reports the results in terms of absolute bias for the standard error estimates and mean coverage rate (CR) for nominal 95% Wald-type confidence intervals over simulations. For a generic parameter estimate $\hat{\theta}_s \in \hat{\boldsymbol{\theta}}$, $s = 1, \ldots, S$, this is computed as

$$\text{CR}(\theta_s) = \frac{1}{B} \sum_{b=1}^{B} \mathbb{I}\left(|\hat{\theta}_s - \theta_0| \leq 1.96 \times \widehat{se}(\hat{\theta}_s) \right), \quad s = 1, \ldots, S,$$

Table 2 Mean (median) ARI across samples for $G = 4$, $D = 3$, and varying N and R

		$R = 20$		$R = 30$	
		Sending	Receiving	Sending	Receiving
N	100	0.64 (0.65)	0.38 (0.40)	0.78 (0.79)	0.49 (0.53)
	500	0.69 (0.71)	0.48 (0.48)	0.83 (0.83)	0.63 (0.67)
	1000	0.70 (0.72)	0.49 (0.51)	0.84 (0.84)	0.63 (0.68)

Table 3 Absolute bias and MSE values across samples for $\boldsymbol{b}$, $\boldsymbol{\mu}$, and $\boldsymbol{\beta}$ for $G = 3$ and $D = 2$, and varying N and R

		$R = 20$		
Absolute bias		$\boldsymbol{b}$	$\boldsymbol{\mu}$	$\boldsymbol{\beta}$
N	100	0.18 0.00 0.01	0.03 0.00	0.17 0.24 0.17 0.32
	500	0.02 0.00 0.00	0.01 0.00	0.12 0.12 0.00 0.02
	1000	0.04 0.00 0.01	0.04 0.00	0.11 0.10 0.01 0.02
MSE		$\boldsymbol{b}$	$\boldsymbol{\mu}$	$\boldsymbol{\beta}$
N	100	0.25 0.00 0.09	0.03 0.00	0.62 0.43 0.14 0.15
	500	0.05 0.00 0.01	0.02 0.00	0.09 0.07 0.02 0.02
	1000	0.03 0.00 0.01	0.02 0.00	0.08 0.06 0.01 0.01
		$R = 30$		
Absolute bias		$\boldsymbol{b}$	$\boldsymbol{\mu}$	$\boldsymbol{\beta}$
N	100	0.02 0.00 0.01	0.01 0.00	0.19 0.26 0.10 0.15
	500	0.02 0.00 0.01	0.02 0.00	0.00 0.01 0.01 0.01
	1000	0.00 0.00 0.01	0.00 0.00	0.03 0.03 0.02 0.02
MSE		$\boldsymbol{b}$	$\boldsymbol{\mu}$	$\boldsymbol{\beta}$
N	100	0.12 0.00 0.06	0.03 0.00	0.34 0.27 0.11 0.10
	500	0.02 0.00 0.01	0.01 0.00	0.05 0.05 0.02 0.02
	1000	0.01 0.00 0.01	0.00 0.00	0.02 0.02 0.01 0.01

Table 4 Absolute bias and MSE values across samples for $\boldsymbol{b}$, $\boldsymbol{\mu}$, and $\boldsymbol{\beta}$ for $G = 4$ and $D = 3$, and varying N and R

		$R = 20$		
Absolute bias		$\boldsymbol{b}$	$\boldsymbol{\mu}$	$\boldsymbol{\beta}$
N	100	0.44 0.00 0.06 0.18	0.17 0.00 0.09	0.23 0.16 0.25 0.20 0.05 0.1
	500	0.33 0.00 0.03 0.08	0.08 0.00 0.34	0.18 0.01 0.16 0.08 0.08 0.05
	1000	0.32 0.00 0.00 0.07	0.05 0.00 0.40	0.13 0.07 0.12 0.01 0.02 0.06
MSE		$\boldsymbol{b}$	$\boldsymbol{\mu}$	$\boldsymbol{\beta}$
N	100	0.56 0.00 0.16 0.30	0.25 0.00 0.27	1.54 0.98 1.03 0.50 0.39 0.24
	500	0.27 0.00 0.07 0.17	0.22 0.00 0.40	0.20 0.17 0.19 0.04 0.05 0.05
	1000	0.13 0.00 0.01 0.04	0.08 0.00 0.38	0.12 0.07 0.06 0.02 0.02 0.01
		$R = 30$		
Absolute bias		$\boldsymbol{b}$	$\boldsymbol{\mu}$	$\boldsymbol{\beta}$
N	100	0.51 0.00 0.11 0.01	0.09 0.00 0.15	0.12 0.02 0.16 0.12 0.04 0.09
	500	0.14 0.00 0.03 0.08	0.02 0.00 0.24	0.06 0.05 0.02 0.03 0.04 0.00
	1000	0.15 0.00 0.02 0.00	0.02 0.00 0.29	0.09 0.05 0.05 0.06 0.06 0.03
MSE		$\boldsymbol{b}$	$\boldsymbol{\mu}$	$\boldsymbol{\beta}$
N	100	0.32 0.00 0.11 0.21	0.12 0.00 0.15	0.81 0.60 0.52 0.23 0.25 0.20
	500	0.11 0.00 0.07 0.11	0.08 0.00 0.18	0.09 0.08 0.08 0.03 0.04 0.03
	1000	0.10 0.00 0.06 0.10	0.03 0.00 0.17	0.06 0.06 0.03 0.03 0.04 0.01

Table 5 Absolute bias for the standard error estimates and CRs for $\boldsymbol{b}$, $\boldsymbol{\mu}$, and $\boldsymbol{\beta}$ when $N = 100$, $R = 20$, $G = 3$, and $D = 2$

	$R = 20$		
	$\boldsymbol{b}$	$\boldsymbol{\mu}$	$\boldsymbol{\beta}$
Absolute Bias	0.005 0.020 0.011	0.001 0.000	0.061 0.002 0.092 0.018
CR	0.93 0.94 0.95	0.96 0.95	0.92 0.94 0.97 0.97

where $\mathbb{I}(\cdot)$ denotes the indicator function, θ_0 denotes the true value of model parameter, and $\widehat{se}(\hat{\theta}_s)$ denotes the standard error of $\hat{\theta}_s$. By looking at this table, it is evident that the bias for the standard error estimates is always under control, and CRs are all very close to the nominal 0.95 confidence level. These results suggest a strong performance of the proposed approach, even when dealing with bipartite networks of a limited size. As stated in the previous sections, we expect that by increasing the number of sending and receiving nodes, results further improve.

Additionally, to evaluate the performance of the proposed model selection strategy, for each simulated network, we estimate the model by letting the number of components and segments vary, $G = 2, 3, 4$ and $D = 1, 2, 3$, and then retain the optimal model by selecting the combination $[G, D]$ providing the minimum BIC value. Results reported in Table 6 show that the correct model (i.e., the one with $G = 3$ and $D = 2$) is selected 98% of the times. Clearly, in more complex settings characterized by a higher number of biclusters, performance is expected to deteriorate. We do still expect that this worsening alleviates with larger sample sizes, emphasizing the crucial role of sufficient data to preserve model accuracy.

5 Application to the Pediatric Patients Bipartite Network

In this section, we analyze the bipartite network entailing connections between pediatric patients with suspected appendicitis and their clinical conditions (Marcinkevičs et al., 2023). In detail, in Section 5.1, we describe the data set. In Section 5.2, we analyze the data using the MLTA model as presented by Gollini and Murphy (2014). In Section 5.3, we apply the proposed biclustering approach, illustrating how it provides a more comprehensive understanding of the phenomenon of interest compared to the original MLTA.

5.1 The Network

The bipartite network considered in the analysis is built starting from a data set that collects information from a cohort of 782 pediatric patients with abdominal pain and suspected appendicitis, admitted to St. Hedwig Children's Hospital in Regensburg (Germany), between 2016 and 2021. For each patient, we have information on the results of ultrasound (US) images, laboratory tests, and clinical scores, for a total of 54 clinical conditions. By excluding

Table 6 Proportion of times the lowest BIC value occurred for varying G and D

G \ D	1	2	3
2	0.00	0.00	0.00
3	0.00	0.98	0.00
4	0.00	0.02	0.00

Table 7 Description of clinical conditions

Clinical condition	Description
Alvarado score	Score based on symptoms, signs, and laboratory test. A score between 0 and 3 corresponds to unlikely appendicitis, 4 to 6 to possible appendicitis, and a score between 7 and 10 to very likely/certain appendicitis.
Pediatric appendicitis score	Score based on patient's history and examination. A score of 5 or less rules out appendicitis, while a score of 6 or more makes a true case of appendicitis highly probable.
Appendix on US	Detectability of the vermiform appendix during sonographic examination. The variable assumes value 1 if detected and 0 otherwise.
Migratory pain	Abdominal pain from epigastrium to the right lower quadrant.
Lower right abdominal pain	Right iliac fossa pain detected on palpation.
Contralateral rebound tenderness	Pain of the contralateral side during the release of pressure over the abdomen.
Coughing pain	Abdominal pain by forced cough.
Nausea	Feeling of sickness/ejection of contents from the stomach through the mouth.
Loss of appetite	Lack or reduction of appetite.
Body temperature	Temperature measured by a thermometer placed in the rectum or in the auditory canal.
WBC count	Number of leucocytes in a unit volume of blood. The normal number is 4500 to 11,000 WBCs per microliter.
Neutrophil percentage	Mature WBC in the granulocytic series. A normal result is between 40 and 60%.
Neutrophilia	Relative neutrophilic leucocytosis, often a result of a bacterial infection.
RBC count	Number of erythrocytes in a unit volume of blood. The standard value for children is between 4.0 and 5.5 million red blood cells per microliter of blood.
Hemoglobin	Hemoglobin level. Standard value for children is between 10 and 15 g/dL.
RDW	A blood test that measures the differences in the volume and size of the erythrocytes. The normal RDW in children is between 12.3 and 14.1.
Thrombocyte count	Number of platelets in a unit volume of blood. In healthy pediatric subjects, normal count ranges between 250,000 and 450,000 μL.
CRP	Protein produced by the liver. Standard values are no higher than 10 mg/L.
Dysuria	Pain or other difficulty during urination.
Stool	Characteristics of bowel movements.
Peritonitis	Spasm of abdominal wall muscles detected on palpation.
Psoas sign	Abdominal pain produced by extension of the hip.
Free fluids	Free fluids inside the abdomen.

missing data, we reduce the data matrix to 542 rows (the sending nodes/patients) and 23 columns (the receiving nodes/clinical conditions). Table 7 shows a detailed description of such conditions.

To construct a binary incidence matrix (and therefore the bipartite network), as described in Section 2, quantitative and categorical variables are transformed into a suitable number of dummy variables, in a coherent manner with respect to the interpretation provided in Table 7. In detail, binarization is carried out as follows:

- Alvarado score: value 1 is assigned if the score is greater than 3, which indicates possible/certain appendicitis.
- Pediatric appendicitis score: value 1 is assigned if the score is greater than or equal to 6, which indicates a possible appendicitis case.
- Body temperature: value 1 is assigned if the body temperature is greater than 37.36.
- WBC count: value 1 is assigned if the count is greater than 11 or lower than 4.5, which corresponds to anomalous values.
- Neutrophil percentage: value 1 is assigned if the percentage is greater than 60 or lower than 40, which corresponds to anomalous values.
- RBC count: value 1 is assigned if the count is greater than 5.5 or lower than 4, which corresponds to anomalous values.
- Hemoglobin: value 1 is assigned if the value is greater than 15 or lower than 10, which corresponds to anomalous values.
- RDW: value 1 is assigned if the value is greater than 14.1 or lower than 12.3, which corresponds to anomalous values.
- Thrombocyte count: value 1 is assigned if the count is greater than 450 or lower than 250, which corresponds to anomalous values.
- CRP: value 1 is assigned if the value is greater than 10, which corresponds to anomalous values.
- Stool: value 1 is assigned if the patient has no diarrhea or constipation.

Overall, a connection between a pair of nodes does exist if the clinical condition is manifested by the patient. The resulting incidence matrix is represented in Fig. 3.

For a deeper understanding, we report in Fig. 4 the distribution of clinical conditions across patients. From this figure, it is evident that most patients have detectable vermiform appendix during sonographic examination (appendix on US) and show a high Alvarado score, lower right abdominal pain, and an anomalous neutrophil percentage. In addition, we can see a rather uniform distribution regarding the pediatric appendicitis score, the loss of appetite, the body

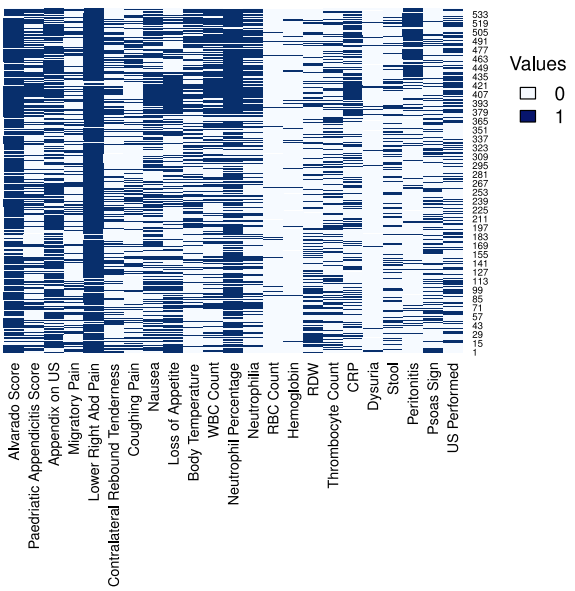


Fig. 3 Binary incidence matrix

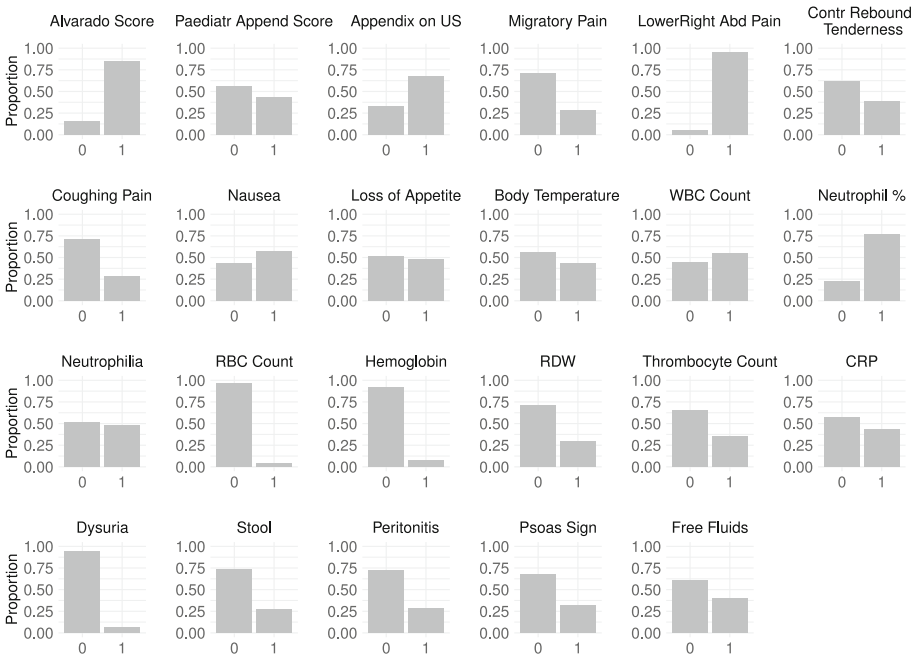


Fig. 4 Distribution of clinical conditions after binarization

temperature, the WBC count, and the neutrophilia. On the contrary, it is evident that more than half of the patients have no migratory pain, contralateral rebound tenderness, coughing pain, abdominal pain produced by extension of the hip (Psoas sign), dysuria, gastrointestinal problems, peritonitis, and free fluids inside the abdomen. In addition, most patients have normal RBC count, hemoglobin, RDW, thrombocyte count, and CRP values.

As regards patients' characteristics, the following nodal attributes are considered: age, BMI (i.e., patient's weight divided by the square of the height), sex, height, weight, length of stay in the hospital, management (i.e., management of the patient assigned by a senior pediatric surgeon – "conservative", "primary surgical", or "secondary surgical"), severity ("uncomplicated", "complicated"). A description of such attributes is shown in Fig. 5. From this Figure, we may see that patients admitted to the hospital are mostly over 10 years old and with a BMI centered around 15. As regards patient's height and weight, the averages are about 149 cm and 44 kg, respectively. In addition, the distribution of patients in terms of gender is homogeneous, and almost all admissions lasted less than 4 days. Finally, it can be noted that most patients were not advised to have surgery and were diagnosed with low symptom severity.

5.2 Analysis of the Network via the MLTA Specification

In this section, the bipartite network described in the previous section is analyzed via the original MLTA model by Gollini and Murphy (2014) with the aim of identifying subsets of patients presenting similar clinical conditions. To this aim, the MLTA is estimated for a varying number of components, G, and latent trait dimensions, D, both ranging from 1 to 4. Both the heterogeneous and the homogeneous specifications (based on component-specific or constant influence parameters, respectively) are considered. To avoid local maxima

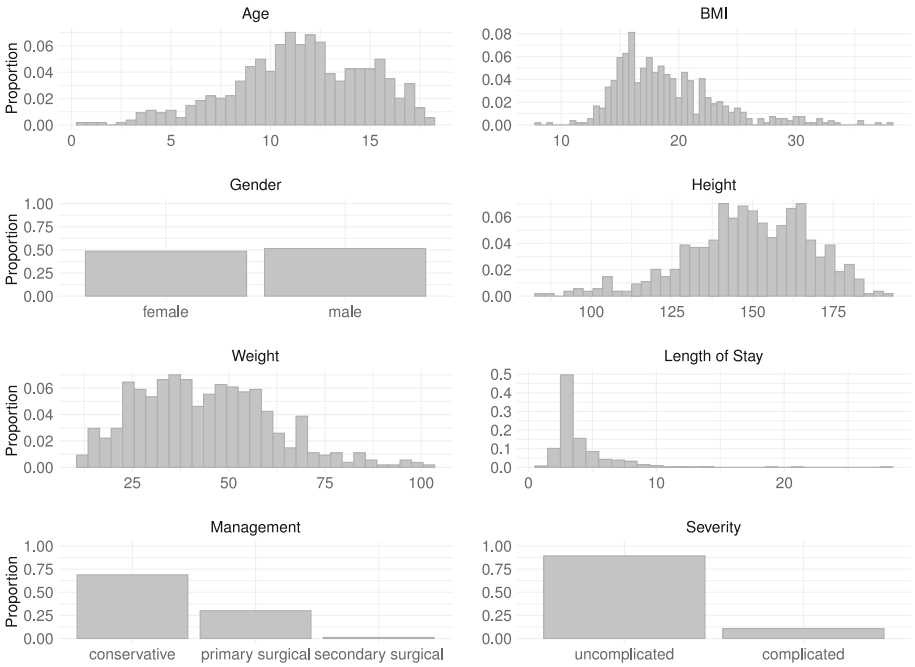


Fig. 5 Distribution of nodal attributes

solutions, a multi-start strategy based on 100 random starts is considered. Table 8 reports the BIC values for all the estimated models; the optimal specification is the one providing the smallest BIC and corresponds to $G = 2$ components, a uni-dimensional latent trait ($D = 1$), and homogeneous influence parameters ($\boldsymbol{w}_{gk} = \boldsymbol{w}_k$).

Figure 6 shows the distribution of the estimated probabilities $\hat{\pi}_{gk}(\hat{\boldsymbol{u}}_i)$ obtained under the selected MLTA specification. These probabilities are derived by substituting into Eq. 1 the

Table 8 BIC for varying numbers of components, G, and latent trait dimensions, D, for the original MLTA model

	Heterogeneous influence parameters			
G \ D	1	2	3	4
1	13105.57	12955.80	13055.33	13296.48
2	13013.13	13146.63	13383.39	13709.73
3	13114.51	13398.86	13758.52	14241.00
4	13262.14	13679.21	14168.83	14738.84
	Homogeneous influence parameters			
G \ D	1	2	3	4
1	13105.56	12955.72	13058.00	13285.45
2	**12892.10**	12948.94	13061.50	13264.69
3	12916.93	12985.75	13110.23	13307.29
4	12962.02	13068.29	13193.25	13363.65

The selected model is highlighted in bold

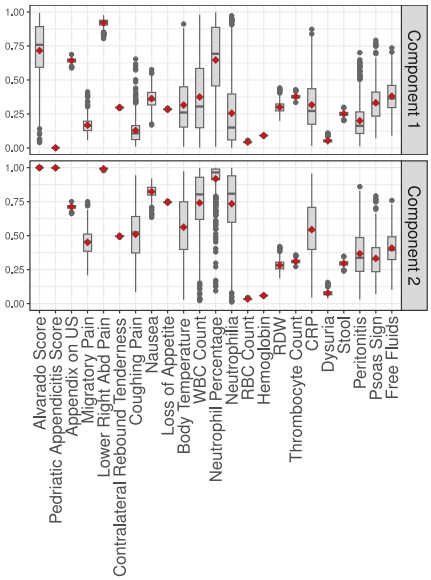


Fig. 6 Distribution of the estimated probabilities $\hat{\pi}_{gk}(\hat{\boldsymbol{u}}_i)$ for $g = 1, 2$ and $k = 1, \ldots, 23$, for the original MLTA. Mean values are highlighted in red

estimates for the attractiveness and the influence parameters, along with the predicted patient-specific latent traits. These latter correspond to the latent traits' posterior means, obtained as $\hat{\boldsymbol{u}}_i = \mathbb{E}[\boldsymbol{u}_i \mid \boldsymbol{y}_i] = \int_{\mathbb{R}^D} \boldsymbol{u}_i f(\boldsymbol{u}_i \mid \boldsymbol{y}_i) d\boldsymbol{u}_i$.

Looking at this figure, we may recognize that patients in the first component have a generally lower chance of manifesting all the clinical conditions under investigation compared with patients classified in component 2. However, it seems that the model does not allow to completely capture similarities/dissimilarities among patients, indicating that there is room for improvement in the data partitioning. This potential is also confirmed by the graphical representation in Fig. 7, where we present the ordered data matrix based on the sending node's partitioning. As is evident, the original MLTA specification misses some relevant patterns in the data, suggesting that enhancements in the data partitioning process could yield better fit and insights. In the subsequent section, we present the analysis of the network via the proposed biclustering procedure.

5.3 Analysis of the Network via the Proposed Biclustering Approach

In this section, we analyze the bipartite network described so far via the proposed biclustering approach to obtain a joint clustering of patients and clinical conditions. More specifically, our goal is to identify blocks in the incidence matrix formed by groups of patients who share specific subsets of clinical conditions. As highlighted above, these patterns may not be detected using the MLTA approach or, more broadly, through traditional clustering methods. In fact, patients may exhibit similarities under certain clinical conditions only, which makes it important to identify these specific patterns. Moreover, taking into account how patients' features influence the probability of belonging to a particular component, we can refine our understanding to gain better insights into clinical phenomena and make more targeted therapeutic decisions.

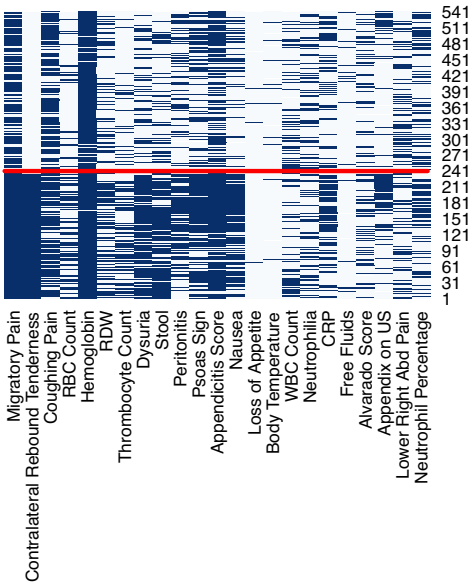


Fig. 7 Ordered data matrix according to the original MLTA

The proposed model is estimated for a number of components, G, and segments, D, both ranging from 1 to 4. To prevent the estimation algorithm from remaining trapped in local maxima, the multi-start strategy described in Section 3.1.2 and based on 100 random starts is considered. As regards the nodal attributes affecting the latent layer of the model, those described in Section 5.1 are considered for the specification of the component prior probabilities $\eta(\boldsymbol{x}_i; \boldsymbol{\beta}_g)$.

Table 9 reports the BIC values for the estimated models; the optimal specification (obtained by looking for the smallest BIC value) corresponds to $G = 2$ components (as for the original MLTA model discussed in the previous section) and $D = 3$ segments.

Estimates for the vector of parameters $\boldsymbol{b} = (b_1, \ldots, b_G)'$ and $\boldsymbol{\mu} = (\mu_1, \ldots, \mu_D)'$ under the optimal specification are $\hat{\boldsymbol{b}} = (-0.59, 0.73)'$ and $\hat{\boldsymbol{\mu}} = (-2.08, -0.18, 1.69)'$, respectively. These results reveal the existence of two different groups of patients characterized by a more (respectively, less) pronounced symptomatology, together with three groups of clinical conditions, being different in the way they are widespread among patients. In detail, the first, the second, and the third segment group together those conditions that are seldom, mildly, and frequently manifested by the patients admitted to hospital, respectively. Overall, the model effectively identifies six homogeneous biclusters.

Table 9 BIC for varying numbers of components, G, and segments, D, for the proposed extended MLTA model

G \ D	1	2	3	4
1	17096.48	15309.17	14805.60	14895.72
2	17251.51	15053.07	**14690.77**	14810.81
3	17444.27	15144.50	14954.31	15183.33
4	17658.37	15205.96	15375.57	15704.42

The selected model is highlighted in bold

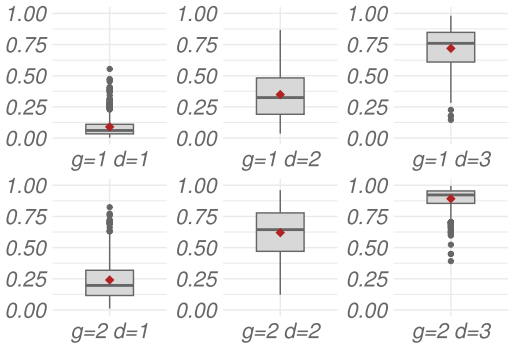


Fig. 8 Distribution of the estimated probabilities $\hat{\pi}_{gk}(\hat{\boldsymbol{u}}_i)$ for $g = 1, 2$ and $d = 1, \ldots, 3$, for the proposed extended MLTA model. Mean values are highlighted in red

A clearer picture of these biclusters can be obtained by looking at the distribution of the estimated probabilities $\hat{\pi}_{gk}(\hat{\boldsymbol{u}}_i)$ reported in Fig. 8. These are derived by substituting into Eq. 2 the estimates of $\boldsymbol{b}$, $\boldsymbol{\mu}$, $\boldsymbol{a}_{gk}$, for $g = 1, 2$ and $k = 1, \ldots, 23$, and $\boldsymbol{u}_i$. As before, $\hat{\boldsymbol{u}}_i$ corresponds to the posterior mean of the latent trait $\boldsymbol{u}_i$. Looking at these distributions, we may recognize that patients in the first component have a low chance of manifesting clinical conditions classified in segment 1 ($\bar{\pi}_{11}$=0.09) with a reduced variability around this center. On the other side, for these patients, the chance of presenting symptoms belonging to segment 2 and 3 is medium-low ($\bar{\pi}_{12}$=0.35) and medium-high ($\bar{\pi}_{13}$=0.72), respectively. Furthermore, heterogeneity between patients is here more pronounced than in segment 1. When focusing the attention on patients classified in the second component, we may observe that they have a medium-low ($\bar{\pi}_{21}$ =0.24), medium-high ($\bar{\pi}_{22}$=0.62), and high ($\bar{\pi}_{23}$=0.89) chance of manifesting clinical conditions belonging to segment 1, 2, and 3, respectively. As far as the heterogeneity captured by the latent trait, we may see from Fig. 8 that this has a low impact on those symptoms belonging to the latter segment.

Table 10 shows the classification of clinical conditions across components. From this table, we may notice that those conditions traditionally attributable to appendicitis (Alvarado score, appendix on US, lower right abdominal pain, and neutrophil percentage) are classified in segment 3, regardless the patient component membership. That is, they are generally more likely to be manifested. Other clinical conditions potentially attributable to appendicitis (pediatric appendicitis score, nausea, loss of appetite, body temperature, WBC count, neutrophilia, CRP, peritonitis, and free fluids) are classified in the second segment, while other more general symptoms (dysuria, stool, RBC count, hemoglobin, RDW, thrombocyte count, migratory pain, contralateral rebound tenderness, coughing pain, Psoas sign) are classified in the first segment for patients presenting more severe symptoms (second component). On the other side, for patients belonging to component 1 and manifesting a milder symptomatology, general symptoms, such as body temperature, neutrophilia, RBC count, hemoglobin, CRP, dysuria, and peritonitis, are all classified in the first segment and are therefore rather unlikely to be observed.

For a clearer understanding, we report in Figs. 9a and b the ordered data matrix according to sending and receiving nodes' partitions and the predicted probabilities $\hat{\pi}_{gk}(\hat{\boldsymbol{u}}_i)$, respectively. These figures clearly demonstrate a substantial improvement in the results compared to

Table 10 Clinical conditions' classification

	Component 1	Component 2
Alvarado score	3	3
Pediatric appendicitis score	2	2
Appendix on US	3	3
Migratory pain	2	1
Lower right abd pain	3	3
Contralateral rebound tenderness	2	1
Coughing pain	2	1
Nausea	2	2
Loss of appetite	2	2
Body temperature	1	2
WBC count	2	2
Neutrophil percentage	3	3
Neutrophilia	1	2
RBC count	1	1
Hemoglobin	1	1
RDW	2	1
Thrombocyte count	2	1
CRP	1	2
Dysuria	1	1
Stool	2	1
Peritonitis	1	2
Psoas sign	2	1
Free fluids	2	2

those presented in Section 5.2. The proposed approach effectively identifies homogeneous blocks within the data matrix, uncovering hidden patterns that were previously undetected by the competing method. This significant advancement highlights the value of the approach, suggesting it is a promising option for data partitioning. As such, the proposal merits serious consideration in practical applications, where a refined understanding of the data is crucial.

To conclude, Table 11 shows the estimated $\boldsymbol{\beta}_2$ coefficients, together with the corresponding 95% confidence intervals, measuring the impact of concomitant variables on patients' clustering. Note that, in this application, the first component is the reference, meaning that $\boldsymbol{\beta}_1 = \mathbf{0}$. Therefore, the estimated coefficients measure the influence of patients' characteristics on the probability of belonging to the second (more affected) component with respect to the first one. From Table 11, it is evident that such a probability is higher for males and for primary and secondary surgical patients. In contrast, the probability decreases with BMI, and it is lower for patients with uncomplicated severity. Age, height, weight, and length of stay do not seem to have a statistically significant effect on the probability of component membership.

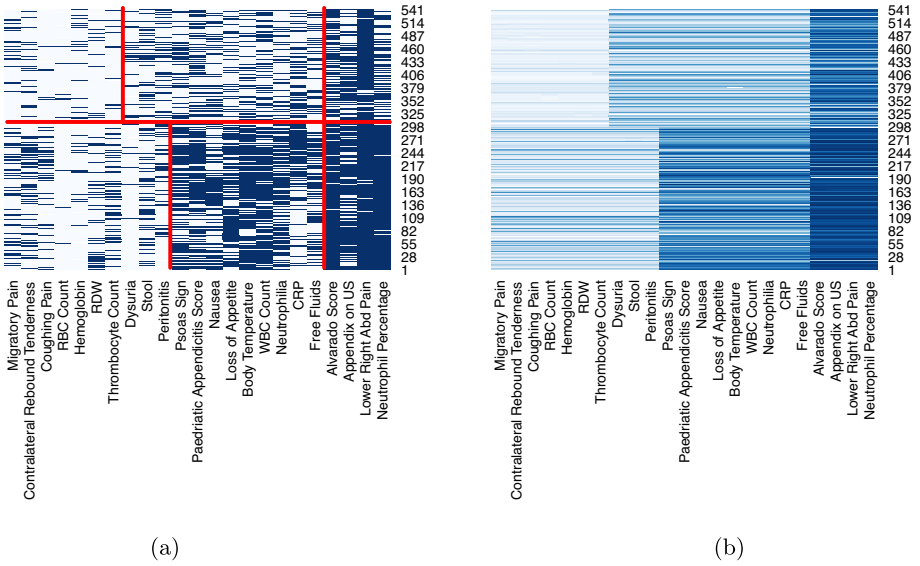


Fig. 9 Ordered data matrix (**a**) and predicted probabilities $\hat{\pi}_{gk}(\hat{\boldsymbol{u}}_i)$ (**b**)

6 Conclusions

In this paper, we extend the MLTA with concomitant variables to perform a joint clustering of sending and receiving nodes in a binary bipartite network. In detail, sending nodes are partitioned into clusters called components, and, in each of them, receiving nodes are partitioned into clusters called segments. Furthermore, the continuous latent trait considered in the model allows us to capture potential residual heterogeneity between sending nodes in the way they connect to receiving nodes. The simulation study we conducted shows that the model can be effectively employed for dimensionality reduction purposes and to identify hidden patterns in a binary incidence matrix. When the number of sending and receiving nodes increases,

Table 11 $\boldsymbol{\beta}_2$ estimates and 95% confidence intervals for the proposed extended MLTA model

	Estimate	Confidence interval
Intercept	12.87	(10.56, 15.18)
Age	-0.05	(-0.20, 0.10)
BMI	-0.26	(-0.45, -0.07)
Male	0.93	(0.88, 0.99)
Height	-0.04	(-0.13, 0.05)
Weight	0.06	(-0.03, 0.14)
Length of stay	-0.10	(-0.44, 0.23)
Management primary surgical	3.97	(3.48, 4.46)
Management secondary surgical	4.04	(1.61, 6.47)
Severity uncomplicated	-4.04	(-5.95, -2.12)

the proposal is more and more able to correctly identify the model parameters and the classification improves. Additionally, simulation results show the ability of the proposed model selection strategy to correctly identify the optimal number of components and segments, along with the good performance of the proposed approach for evaluating uncertainty of parameter estimates. The model is also applied to the bipartite network entailing connections between pediatric patients with suspected appendicitis and their clinical conditions, with the aim of identifying blocks of patients sharing similar subsets of clinical conditions. In detail, the proposed model identifies 2 components of patients according to their symptomatology and 3 segments of conditions based on their diffusion among patients. Furthermore, the analysis shows that the patients' clustering is influenced by BMI, gender, management, and severity assigned by the doctor. The empirical example is used to compare the conventional MLTA method with the proposed extension with the aim of further emphasizing the improvements made by incorporating the biclustering approach.

Both the simulation study and the real data analysis are carried out via the open-source R software (R Core Team, 2024). All the codes and the data used in this paper are available in the GitHub repository https://github.com/dfailli/MLTA_biclustering.

It is worth highlighting that, besides the proposal is detailed for dealing with binary bipartite network, it can be effectively applied also out of the network field, to deal with general binary data matrix. As far as methodological advances of the proposed approach, various extensions are possible. To begin, the proposed model specification assumes homogeneity of the latent traits, imposing unit variance across components and variables. While this simplifies the model, it may be overly restrictive in certain contexts. A key area for further development involves relaxing these constraints, a topic that is currently under investigation. In addition, the model can be modified to handle response variables with more than two categories, as well as to deal with longitudinal bipartite networks, by allowing for a dynamic joint clustering of sending and receiving nodes, where nodes may move from one partition to another over time.

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Data Availability The data set is available at the web page https://doi.org/10.5281/zenodo.7669442.

Code Availability The code is available in the first author's GitHub repository https://github.com/dfailli/MLTA_biclustering.

Declarations

Conflict of Interest The authors declare no competing interests.

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